

Backend Automated Test Suite Calculation Logic & Tenant Isolation

Repo: wergonic-django-backend · **Type:** Epic (split into sub-tasks) · **Owner:** _____ · **Date:** 2026-08-03 · **Rev 2** (CEO-reconciled)

~510

TOTAL TEST CASES

~205

P0 MUST-SHIP

22

AUDIT BUGS MAPPED

42

MODULES COVERED

Context

The backend currently has ~17 real test methods across 14 apps (most `tests.py` files are empty stubs) and no test infrastructure. CI runs `manage.py test`, finds almost nothing, and reports green — so "green" today means "nothing is tested," not "nothing is broken." Meanwhile the two 2026-07 audits surfaced concrete calculation errors and tenant-isolation (cross-org data leak) defects. This epic builds a real suite, **starting from the code we already know is fragile**.

Scope produced by inventorying all 42 logic-bearing modules and cross-referencing both audit docs (35 findings; 22 unit-testable). Every cited bug site was verified in source.

Objective

- Stand up backend test infrastructure (none exists today).
- Lock down the audited calculation + permission defects with tests (regression protection as we fix).
- Establish a baseline suite that CI actually gates on, following the Flutter repo's standard (78 files / ~498 tests).

✓ Definition of Done (per sub-task)

- Listed module has tests for its main path **plus** the specific edge/boundary cases noted.
- Each mapped audit finding has a test that fails on the buggy version and passes on the fixed one.
- Suite runs green in CI (`pytest`), wired into the GitHub Actions workflow.
- No test touches the real staging/production DB, Firebase, or live data — tests run against isolated test data only.

Reading the scope

The `est` column is a rough count of *logical test cases* per target — a relative sizing signal, not a schedule. The **P0 slice (~205 cases) is the must-ship**: it covers every audited defect. Sequencing/estimation is the team's call.

🔄 Rev 2 — reconciled with the CEO calc-issue review (2026-08-03)

Expected values below follow Farhad Abtahi's rulings, verified against the code:

- **Head** backward-bending threshold → -5° (was -10°). The real numeric flip lives in `risk_assessment_posture` (the risk verdict) + the Word report — see **2.18** — *not* `ramp_calculator`, which already uses -5° and only needs its stale `-10` labels renamed.
- **Trunk** backward-bending **risk verdict stays -10°** (deliberate). Do not conflate the two.
- **TNO** (`extractTaskCharacteristics`) dropped from P0 — CEO: "skip TNO for now" → moved to backlog.
- **Score-1.5** (P8) confirmed a real bug — `calculate_postures_2`, see **1.2**.
- **Overlap-criteria wording** (P16) is *not* a bug — `posture_thresholds` stays as a pure-logic test, not a fix.

Sub-task 0 · Test infrastructure — PREREQUISITE

There is no test infrastructure today (only a bare test runner, no reusable fixtures). This must exist before any database- or view-level test below can run — the approach and tooling are the dev's call.

- A working test-run setup that CI executes and gates on (today CI runs a near-empty step that reports green).
- Reusable data fixtures/factories for the core models the tests need: User, Organization, UserOrganization, Session, Measurement, MeasurementData, Event, Category, Label, SessionActivity, SessionCondition, WorkCycleEvent, RAMPAssessment, SessionStats, Ticket, Notification.

Phase 1 · P0 pure-calculation logic — DO FIRST (~42 cases)

Highest value-per-hour: pure functions, call with dicts/lists/DataFrames, no DB. Kills the worst calc audit findings fast.

#	Module	est	Audit	Key cases
1.1	<code>mec_mapping.py :: calculate_time</code>	8	mechanics	Generic threshold-agnostic band accumulator — no $-5/-10$ baked in (callers pass the threshold). Upper-INCLUSIVE (<code>v≤supper</code>), lower-exclusive (<code>v>lower</code>); band <code>(20,45]</code> : 20 excluded, 45 included, 45.1 excluded; <code>side_bending</code> uses <code>abs()</code> ; <code>elevation_angle</code> drops <code>v<0</code> ; %=time-weighted (matching_ms/valid_ms); empty→{0,0,0}
1.2	<code>mec_mapping.py :: calculate_postures_1..6</code>	18	P8	CONFIRMED BUG <code>calculate_postures_2</code> emits 1.5 (input band [1,6.25)). Pre-fix: reproduce 1.5 (red). Post-fix: never returns 1.5, scores monotonic in %; don't hard-code band values until the fix is defined. None→None; clamp <code><0→0, >100→top</code> ; other ladders (1,3,4,5,6) never emit 1.5
1.3	<code>mec_segments.py :: scale_results_with_takt_time, get_feedback_frequency</code>	6	supports A15	takt=60,count=10,static=120 → 600/120; takt≤0 raises; freq: metadata None→52 hand / 13 else; arm variants fall to standard branch
1.4	<code>posture_thresholds.py</code> (3 fns)	10	P16 not a bug	Pure threshold tables that feed every report. CEO: the overlap wording (P16) is fine — test as pure logic, not a fix. Each boundary in exactly one band; ranges end <code>(100,'high')</code> ; integers render without <code>.0</code>

Phase 2 · P0 tenant isolation + integration (~163 cases)

The cross-org data-leak blockers. Test the **shared gate AND the callers** — the audit fixes all delegate to a couple of shared functions, which are the real single points of failure.

#	Module	est	Audit	Note
2.1	<code>users/scoping.py :: get_request_organization</code>	8	linchpin A1/A3/ A5/A6/ A7	ADDED · CRITICAL ~15 endpoints delegate here. X-Client-Platform: web → consulting vs active org; _cached_org memoization; unauth short-circuit. Wrong org here = every "fixed" view silently breaks while its test still passes.
2.2	<code>user_sessions/permissions.py</code> (3 classes)	12	session gates	ADDED The control the A1 target assumes is "already guarded" but never tested. ⚠ Latent bug: <code>SessionCrudPermission</code> does <code>session.worker.user</code> → <code>AttributeError</code> when worker is None (same root as A18). Add a dedicated case.
2.3	<code>work_tags/permissions.py</code> (4 classes)	18	A1 A3	org-admin/ergonomist gates; cross-org → False; <code>is_deactivated=True</code> → False; swagger bypass behavior
2.4	<code>users/permissions/level_0/1/2 + userCrud</code>	20	—	role-based CRUD permission matrix
2.5	<code>work_tags/views.py</code> (Event/Category/Label, APIClient)	12	A1 A3	URL <code>org_id / category_id</code> must not be trusted over the caller's org
2.6	<code>tickets/views.py</code> (integration)	6	A4	tickets scoped to org — no global IDOR
2.7	<code>stats_view.py :: GenerateGeminiAnalysisView.get</code>	5	A5	AI report GET rejects another org's session
2.8	<code>work_library/views.py + serializers.py</code>	6	A6	update can't re-point to another org's activity
2.9	<code>workcycles/views.py :: TaskViewSet.get_queryset</code>	5	A7	org-less orphan tasks not exposed cross-company
2.10	<code>ramp_calculator.py :: run_ramp_calculations</code>	10	P2 A16	Head backward ALREADY uses <code>upper_threshold=-5</code> (assert -5, not -10); after rename, payload has <code>postures_2...gt_5_degrees_*</code> keys, not <code>gt_10</code> ; trunk posture-4 uses -5 (unchanged); emits NO -10 value (anti-conflation — the -10 verdict lives in 2.18); task-scoped posture 6 uses task-window not whole-session (A16)
2.11	<code>mec_segments.py :: compute_mec_for_segment</code>	8	A15	DB-BACKED Trunk with side-bending present but forward-bending null → no swallowed <code>UnboundLocalError</code> , missing-limb flags set
2.12	<code>management/commands/session_check.py</code>	5	A18	per-minute auto-finish doesn't crash on sessions with a deleted worker
2.13	<code>management/commands/archive_sessions.py</code>	6	A19	auto-archive uses post-rename status value
2.14	<code>notifications/models.py + views.py</code>	6	A20	failed-measurement notification filed under session's org, not recipient's active org
2.15	<code>stats_view.py :: single-session Excel export</code>	5	A17	head values not written over trunk columns
2.16	<code>measurements.py :: merge_measurment_files</code>	9	A11	REGRESSION GUARD · already fixed Merged zip includes <code>_metadata_merged.json</code> , highest App.Version chosen, axis-swap correct
2.17	<code>session_views.py :: session merge</code>	12	A10	REGRESSION GUARD · already fixed Measurements rebuilt from combined CSV, not moved. Re-estimated higher: needs the full async merge→reprocess pipeline

#	Module	est	Audit	Note
2.18	<code>risk_assessment_posture.py</code> :: <code>calculate_risk_posture_stats</code> + <code>session_docx_report.py</code>	10	P2 P19	ADDED · the real -5° flip HEAD backward risk verdict flips -10°→-5° (mask ≤-5 : a head row at -6° now counts; id/label say -5°). TRUNK verdict stays -10° (trunk -7° doesn't count; -10°/-12° do). Assert head vs trunk side-by-side; Word-report head row label -5° , trunk -10°

Phase 3 · P1 important logic — backlog (~199 cases)

Not audited, medium risk. Filters, report-math, signal-processing helpers. Grouped counts (full per-module breakdown on request):

Area	Modules	est
Signal processing	<code>mec_mapping.py</code> — detect_peaks, static periods, butter filter, arm postures, zones, hourly	22
Wrist analysis	<code>wrist_mec_mapping.py</code> — position, custom zones, peaks, static	24
Report math	<code>report_service.py</code> — label-bounds parse, bin midpoints, weighted mean/median, option/velocity blocks	46
A16 numeric core ADDED	<code>transformations.py</code> :: <code>compute_distance_zone_minutes</code> (+ round/tz helpers) — pure distance-zone banding feeding posture 6	8
Pie / risk stats	<code>pie_stats.py</code> , <code>risk_assessment_posture.py</code> — P18/P19 banding (guard intentional constants; the head -5° flip itself is P0 → 2.18)	24
Data reshaping	<code>stats.py</code> , <code>measurements.py</code> (pure helpers), <code>task_measurments.py</code> , <code>filters.py</code> :: <code>pair_measurements_by_time</code> , preprocess/swapping	58
DOCX helpers	<code>session_docx_report.py</code> :: <code>get_risk_image</code> , <code>format_duration_seconds</code>	11

Phase 4 · P2 nice-to-have — backlog (~110 cases)

Lower risk / harder to test.

TNO pipeline (incl. `extractTaskCharacteristics` , the old P1 target) — **deferred per CEO: "skip TNO for now"** . Also: inversion correction, time bounds, percentage-change, swapped pie stats, user-service claims, report JSON/excel value maps, work_tags event upsert, and `firebase_auth/authentication.py` :: `authenticate` (token-parsing branch — **ADDED** , security-relevant: silently returns None on unknown uid). Report-formatting findings P7/P17 live here.

Explicitly OUT OF SCOPE (don't spend time here)

- Raw-SQL percentile paths in `stats.py` (Postgres/Timescale-only, huge fixtures) — reusable zone/pivot logic is covered elsewhere.
- Firebase/DB IO orchestration in `measurements.py` , `task_measurments.py` , `calculation_service.py` — underlying math covered by pure targets.
- DOCX/XLSX byte-assembly plumbing (`build_session_word_report` , excel builders, `docx_pie_charts.py`) — data-selection branching covered via helpers.
- Firebase Storage pass-throughs (`tickets/utils.py` , `devices/utils.py`) and `user_service` create/update/delete (heavy multi-system mocking).